

Orthogonality

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Q1

- (i) The two rows are multiples, so the row space has basis

$$\{(1, 2, 1)\}, \quad \dim \mathcal{R}(A) = 1.$$

The columns are also multiples of $(1, 2)^T$, so

$$\{(1, 2)^T\} \text{ is a basis for } \mathcal{C}(A), \quad \dim \mathcal{C}(A) = 1.$$

The equation $Ax = 0$ reduces to

$$x_1 + 2x_2 + x_3 = 0.$$

Thus

$$x = x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix},$$

and a basis for the nullspace is

$$\left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}, \quad \dim \mathcal{N}(A) = 2.$$

Finally,

$$A^T y = 0 \iff y_1 + 2y_2 = 0,$$

so a basis for the left nullspace is

$$\left\{ \begin{bmatrix} -2 \\ 1 \end{bmatrix} \right\}, \quad \dim \mathcal{N}(A^T) = 1.$$

- (ii) The row-space basis vector is perpendicular to both nullspace basis vectors:

$$(1, 2, 1) \cdot (-2, 1, 0) = 0, \quad (1, 2, 1) \cdot (-1, 0, 1) = 0.$$

The column-space and left-nullspace basis vectors satisfy

$$(1, 2) \cdot (-2, 1) = 0.$$

These calculations verify the orthogonality relations for the four fundamental subspaces.

Q2

- (i) A vector $x = (x_1, x_2, x_3, x_4)$ belongs to $S^\perp$ exactly when

$$x \cdot s_1 = x_1 + x_2 = 0, \quad x \cdot s_2 = x_2 + x_3 = 0.$$

Let $x_2 = r$ and $x_4 = s$. Then

$$x = (-r, r, -r, s) = r(-1, 1, -1, 0) + s(0, 0, 0, 1).$$

Therefore a basis for $S^\perp$ is

$$\{(-1, 1, -1, 0), (0, 0, 0, 1)\}.$$

- (ii) Both s_1 and s_2 are perpendicular to the two displayed basis vectors of $S^\perp$. Hence $S \subseteq (S^\perp)^\perp$. The vectors s_1 and s_2 are independent, so $\dim S = 2$. Since $\dim S^\perp = 2$ in $\mathbb{R}^4$,

$$\dim (S^\perp)^\perp = 4 - 2 = 2.$$

The inclusion is therefore between subspaces of the same dimension, and

$$(S^\perp)^\perp = S.$$

Q3

- (i) The projection onto the line through a has the form

$$p = \frac{a^T b}{a^T a} a.$$

Here

$$a^T b = 1 \cdot 2 + 2 \cdot 1 + 2 \cdot 1 = 6, \quad a^T a = 1 + 4 + 4 = 9.$$

Therefore

$$p = \frac{2}{3} a = \begin{bmatrix} 2/3 \\ 4/3 \\ 4/3 \end{bmatrix}, \quad e = b - p = \begin{bmatrix} 4/3 \\ -1/3 \\ -1/3 \end{bmatrix}.$$

- (ii) We have

$$a^T e = 1 \cdot \frac{4}{3} + 2 \left(-\frac{1}{3}\right) + 2 \left(-\frac{1}{3}\right) = 0,$$

so $e \perp a$. Also,

$$\|b\|^2 = 6, \quad \|p\|^2 = 4, \quad \|e\|^2 = 2.$$

Thus $\|b\|^2 = \|p\|^2 + \|e\|^2$.

- (iii) The projection matrix is

$$P = \frac{aa^T}{a^T a} = \frac{1}{9} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{bmatrix}.$$

Since $a^T a = 9$,

$$P^2 = \frac{aa^T aa^T}{(a^T a)^2} = \frac{a(a^T a)a^T}{(a^T a)^2} = P.$$

Q4

- (i) The columns of A are $(1, 0, 1)^T$ and $(0, 1, 1)^T$. We have

$$A^T A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad A^T b = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

The normal equations are

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Since

$$(A^T A)^{-1} = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix},$$

we obtain

$$\hat{x} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad p = A\hat{x} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}.$$

- (ii) The error is

$$e = b - p = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}.$$

Its dot products with the columns of A are

$$(1, 0, 1) \cdot (1, 1, -1) = 0, \quad (0, 1, 1) \cdot (1, 1, -1) = 0.$$

Thus $A^T e = 0$.

- (iii) Multiplying the matrices gives

$$P = A(A^T A)^{-1} A^T = \frac{1}{3} \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}.$$

Q5

- (i) Because $A^T A$ is symmetric, its inverse is symmetric. Therefore

$$P^T = \left(A(A^T A)^{-1} A^T \right)^T = A(A^T A)^{-1} A^T = P.$$

Furthermore,

$$\begin{aligned} P^2 &= A(A^T A)^{-1} A^T A(A^T A)^{-1} A^T \\ &= A(A^T A)^{-1} (A^T A) (A^T A)^{-1} A^T = P. \end{aligned}$$

- (ii) For every b , the vector Pb is of the form Ay , so $Pb \in \mathcal{C}(A)$. Also,

$$A^T(b - Pb) = A^T b - A^T A(A^T A)^{-1} A^T b = 0.$$

Thus $b - Pb \in \mathcal{N}(A^T)$, which is perpendicular to $\mathcal{C}(A)$. Hence Pb is the projection of b onto $\mathcal{C}(A)$.

- (iii) Since $P^T = P$ and $P^2 = P$,

$$(I - P)^T = I - P, \quad (I - P)^2 = I - 2P + P^2 = I - P.$$

For any b , $P(I - P)b = Pb - P^2b = 0$, so $(I - P)b$ is in the nullspace of P . If $A^Tb = 0$, then $Pb = 0$. Conversely, if $Pb = 0$, then $b = (I - P)b$, and

$$A^T(I - P)b = A^Tb - A^T A(A^T A)^{-1} A^T b = 0.$$

Thus the nullspace of P is $\mathcal{N}(A^T)$, and $I - P$ projects onto that subspace.

Q6

(i) The model is

$$A \begin{bmatrix} c \\ d \end{bmatrix} \approx b, \quad A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

The two normal equations are obtained from

$$A^T A = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}, \quad A^T b = \begin{bmatrix} 5 \\ 6 \end{bmatrix}.$$

Thus

$$3c + 3d = 5, \quad 3c + 5d = 6.$$

(ii) Subtracting the first equation from the second gives $2d = 1$, so

$$\hat{d} = \frac{1}{2}, \quad \hat{c} = \frac{7}{6}.$$

The fitted values are

$$p = A \begin{bmatrix} 7/6 \\ 1/2 \end{bmatrix} = \begin{bmatrix} 7/6 \\ 5/3 \\ 13/6 \end{bmatrix}.$$

(iii) The residual is

$$e = b - p = \begin{bmatrix} -1/6 \\ 1/3 \\ -1/6 \end{bmatrix}.$$

The columns of A are $(1, 1, 1)^T$ and $(0, 1, 2)^T$, and

$$(1, 1, 1) \cdot e = 0, \quad (0, 1, 2) \cdot e = 0.$$

This is the normal-equation condition $A^T e = 0$.

Q7

- (i) Set $q_1 = a_1/\|a_1\|$. Then

$$q_1 = \frac{1}{\sqrt{2}}(1, 1, 0).$$

The next orthogonal vector is

$$v_2 = a_2 - (q_1^T a_2)q_1 = \left(\frac{1}{2}, -\frac{1}{2}, 1\right),$$

so

$$q_2 = \frac{1}{\sqrt{6}}(1, -1, 2).$$

For the third vector,

$$v_3 = a_3 - (q_1^T a_3)q_1 - (q_2^T a_3)q_2 = \frac{2}{3}(-1, 1, 1).$$

Therefore

$$q_3 = \frac{1}{\sqrt{3}}(-1, 1, 1).$$

The three vectors are unit vectors and are mutually orthogonal by construction.

- (ii) The entries of $R = Q^T A$ are $r_{ij} = q_i^T a_j$. Thus

$$R = \begin{bmatrix} \sqrt{2} & 1/\sqrt{2} & 1/\sqrt{2} \\ 0 & \sqrt{6}/2 & 1/\sqrt{6} \\ 0 & 0 & 2/\sqrt{3} \end{bmatrix}.$$

Consequently $A = QR$, and R is upper triangular.

Q8

- (i) The two columns of Q are orthonormal, so $Q^T Q = I_2$. The coordinates of the projection are

$$Q^T b = \begin{bmatrix} q_1^T b \\ q_2^T b \end{bmatrix} = \begin{bmatrix} 3/\sqrt{2} \\ 7/\sqrt{6} \end{bmatrix}.$$

Therefore

$$\begin{aligned} p = QQ^T b &= \frac{3}{\sqrt{2}}q_1 + \frac{7}{\sqrt{6}}q_2 \\ &= \begin{bmatrix} 8/3 \\ 1/3 \\ 7/3 \end{bmatrix}. \end{aligned}$$

- (ii) The error is

$$e = b - p = \begin{bmatrix} -2/3 \\ 2/3 \\ 2/3 \end{bmatrix}.$$

Direct calculation gives

$$q_1^T e = 0, \quad q_2^T e = 0,$$

and hence $Q^T e = 0$. Thus the error is perpendicular to the whole column space of Q .